

Decrease and Conquer Algorithms

Consider a class P of problems P parameterized over $\mathbb{N}$ usually by its size, denoted by n , such that we have

1. Base cases: Set P_0 of small-size problems so for any $P \in P_0$, the solution is explicitly given by $s(P)$
2. Reduction step: We have a routine F that reduces any given problem X into a smaller (possibly modified) subproblem A
3. Construction step: We have a routine G that constructs the solution to the bigger problem X given the solution to the smaller subproblem A

General Form

Function $\text{DecC}(P)$:

```

if  $P \in P_0$  then
    return  $s(P)$ 
else
     $A \leftarrow F(P)$ 
     $A' \leftarrow \text{DecC}(A)$ 
    return  $G(A')$ 

```

Divide and Conquer Algorithms

Consider a class P of problems P parameterized over $\mathbb{N}$ usually by its size, denoted by n , such that we have

1. Base Cases: Set P_0 of small-size problems such that for any problem $P \in P_0$, the solution is given explicitly by $s(P)$
2. Reduction Step: We have a routine F that reduces any given problem X into smaller (possibly modified) subproblems A and B
3. Construction Step: We have a routine G that constructs the solution to the bigger problem X , given the solution to smaller subproblems A and B is known

General Form

Function $\text{DivC}(P)$:

```

if  $P \in P_0$  then
    return  $s(P)$ 
else
     $A, B \leftarrow F(P)$ 
     $A', B' \leftarrow \text{DivC}(A), \text{DivC}(B)$ 
    return  $G(A', B')$ 

```

Example

Given a sorted array L (increasing) of n integers and integer x , find whether $x \in L$ (Binary search)

- Base case $\Rightarrow$ Single element array
- Reduction step $\Rightarrow$ Compare x with middle element and reduce to appropriate half
- Construction step $\Rightarrow$ Nothing

Example

Given a 2D map M made of n straight lines, find a proper coloring (adjacent regions have distinct colors) of M with just 2 colors

- Base case $\Rightarrow n = 1$
- Reduction step $\Rightarrow$ Removing the n^{th} line
- Construction step $\Rightarrow$ Switch colors on one side of the line

Example

Given an array L of n integers, sort L in increasing order

- Base case: Single element array
- Reduction step: Bisect list into two halves
- Construction step: Merge two lists while keeping order

Example

Given 3ⁿ coins, all equal weight except one which is fake (heavier), find the fake using minimum weighings

- Base case: 1 coin $\Rightarrow$ must be fake
- Reduction step: Split into 3 stacks $\Rightarrow$ If 2 are equal the coin is in the third etc.
- Construction step: Nothing

Example

Given 2 integers a, b of n digits (in base d), compute their product ab

- Base case: Single digit a, b
- Reduction step: dissect each

$$a = a_0 + a_1 d^{n/2}$$

$$b = b_0 + b_1 d^{n/2}$$

- Construction step:

$$ab = \underbrace{a_0 b_0}_{\frac{n}{2}} + \underbrace{(a_0 b_1 + \underbrace{a_1 b_0}_{\frac{n}{2}})}_{\frac{n}{2}} + \underbrace{a_1 b_1}_{\frac{n}{2}} d^n = \underbrace{a_0 b_0}_{\frac{n}{2}} + \underbrace{a_1 b_1}_{\frac{n}{2}} d^n$$

Connection to Master TheoremAnalysis

Coloring map: $T(n) = T(n-1) + (n-1)$

Binary Search: $T(n) = T(\frac{n}{2}) + 1$

Merge Sort: $T(n) = 2T(\frac{n}{2}) + \Theta(n)$

Fake Coin: $T(n) = T(\frac{n}{3}) + 1$

Karatsuba: $T(n) = 3T(\frac{n}{2}) + \Theta(n)$

- Can use extended Master Theorem to determine Θ class